

Supplementary for Solow Growth Model II

Abstract

In last lecture note, we discussed the population growth rate in the model but did not address technological progress in labor. This supplementary material aims to provide a brief explanation of the Solow model's consideration of technological progress in labor, which is referred to as effective labor. I also provide you with some important concepts about growth model that are not covered in the textbook but professor mentioned in class.

Model

Let $E = AN$ and define $\lambda = \frac{\dot{A}}{A}$ is growth rate of technological progress. We can treat A as the technology of labor and write the homogenous of degree 1 production function as:

$$Y_t = F(AN_t, K_t); \quad F_E > 0, F_K > 0, F_{EE} < 0, F_{KK} < 0, F_{EK} > 0$$

let $\tilde{y} = \frac{Y}{AN}$ is output per effective labor, $\tilde{k} = \frac{K}{AN}$ is capital per effective labor.

$$Y = F(AN, K) \rightarrow \frac{Y}{AN} = F\left(\frac{AN}{AN}, \frac{K}{AN}\right) \rightarrow \tilde{y} = F\left(1, \frac{K}{AN}\right) \rightarrow \tilde{y} = f(\tilde{k})$$

1. $Y = C + S$
2. $I = S$
3. $S = sY$
4. $\dot{K} = I - \delta K$

the fourth relation is the continuous case of $K_{t+1} = (1 - \delta)K_t + I_t$, where $\dot{K} = \frac{\partial K}{\partial t}$ is the continuous form of $K_{t+1} - K_t$.

Combine the above equations, we have

$$\dot{K} + \delta K = sY$$

divide both sides by AN

$$\frac{\dot{K}}{AN} = s \frac{Y}{aN} - \delta \frac{K}{aN} = s\tilde{y} - \delta\tilde{k} = sf(\tilde{k}) - \delta\tilde{k} \quad (1)$$

$$\tilde{k} = \frac{K}{AN} \rightarrow \ln \tilde{k} = \ln K - \ln A - \ln N \quad (2)$$

take partial derivative with respect to t on both sides

$$\frac{\dot{k}}{\tilde{k}} = \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{N}}{N} = \frac{\dot{K}}{K} - \lambda - n; \quad \dot{\tilde{k}} = \frac{\partial \tilde{k}}{\partial t}$$

multiply both sides by $\tilde{k}$

$$\begin{aligned} \dot{\tilde{k}} &= \frac{\dot{K}}{K} \frac{K}{AN} - (n + \lambda)\tilde{k} = \frac{\dot{K}}{AN} - (n + \lambda)\frac{K}{AN} \\ \frac{\dot{K}}{AN} &= sf(\tilde{k}) - \delta\tilde{k} \end{aligned} \quad (3) \quad (\text{from equation (1)})$$

then substitute $\frac{\dot{K}}{AN}$ into $\dot{\tilde{k}} = \frac{\dot{K}}{AN} - (n + \lambda)\frac{K}{AN}$:

$$\dot{\tilde{k}} = sf(\tilde{k}) - (n + \delta + \lambda)\tilde{k} \quad (4)$$

Equation (4) is the **fundamental accumulation equation**, when $\dot{\tilde{k}} = 0$, the economy reaches the steady state. Let's see a numerical example below.

Example 1 (We omitted the time index t for convenience) Consider the production function $Y = K^\alpha (AN)^{1-\alpha}$, we can write the output per effective worker by dividing both sides by AN

$$\tilde{y} = \frac{Y}{AN} = K^\alpha (AN)^{-\alpha} = \left(\frac{K}{AN}\right)^\alpha = \tilde{k}^\alpha$$

Given the technology progress rate, population growth rate, depreciation rate, and saving rate are λ, n, δ , and s , respectively. According to the steady-state condition $\dot{\tilde{k}} = 0$ or $s\tilde{y} = (n + \delta + \lambda)\tilde{k}$, we can derive

$$s\tilde{y} = (n + \delta + \lambda)\tilde{k} \rightarrow s\tilde{k}^\alpha = (n + \delta + \lambda)\tilde{k} \rightarrow \frac{s}{n + \delta + \lambda} = \tilde{k}^{1-\alpha} \rightarrow \tilde{k}^* = \left(\frac{s}{n + \delta + \lambda}\right)^{\frac{1}{1-\alpha}}$$

substitute $\tilde{k}^*$ into $\tilde{y}$, then we can get $\tilde{y}^* = \left(\frac{s}{n + \delta + \lambda}\right)^{\frac{\alpha}{1-\alpha}}$. $\tilde{k}^*$ is the steady state capital per effective worker, $\tilde{y}^*$ is the steady-state output per effective worker. ■

Growth rates of Y_t and K_t when economy reach the steady-state are:

1.

$$\tilde{k} = \frac{K}{AN} \rightarrow \frac{\dot{\tilde{k}}}{\tilde{k}} = \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{N}}{N} = \frac{\dot{K}}{K} - (\lambda + n) \rightarrow \frac{\dot{K}}{K} = n + \lambda$$

2.

$$\tilde{y} = \frac{Y}{AN} \rightarrow \frac{\dot{\tilde{y}}}{\tilde{y}} = \frac{\dot{Y}}{Y} - \frac{\dot{A}}{A} - \frac{\dot{N}}{N} = \frac{\dot{Y}}{Y} - (\lambda + n) \rightarrow \frac{\dot{Y}}{Y} = n + \lambda$$

So, the growth rates of Y and K are equal to the technology progress rate plus population growth rate, which are exogenous variable. This is why the Solow growth model is also called **exogenous growth model**.

Convergence hypothesis

Definition 0.1. *Over time, poorer economies(countries) tend to catch up with richer economies(countries) in terms of income per worker, productivity, and other economic indicators.*

1. Absolute convergence hypothesis: suggests that over time, economies with lower levels of income per worker will experience higher growth rates of output, and thus, eventually converge to the same level of income per worker as economies with higher initial levels of income per worker. **This hypothesis implies that there is a negative relationship between a country's initial income per worker and its subsequent growth rate.** In other words, the poorer the country is at the beginning, the faster its economy will grow in order to catch up with richer countries.

2. Relative convergence hypothesis: The output growth rate per worker in countries with different economic structures **is determined by the relative gap between the current output per worker and the output per worker in their steady-state equilibrium.** If a country's current output per worker is closer to its steady-state equilibrium than another country's, then it will have a lower growth rate. Conversely, if a country's current output per worker is farther from its steady-state equilibrium than another country's, then it will have a higher growth rate.

Endogenous Growth Model

Since the exogenous growth theory fails to reflect the fact of income growth per worker, we must revise the model by incorporating growth rate into it, so that the model can endogenously explain the growth facts. If we want to explain the fact that there are income growth and capital per worker growth for developed countries, we must find a way to avoid the diminishing marginal productivity of capital in the production function. This is the characteristic of endogenous growth theory.

In order to reflect the fact of population stabilization in developed countries, endogenous growth theory assumes a fixed labor force in the model design, with labor normalized to one unit. Therefore, there is no population growth rate, that is, $n = 0$.

AK Model

The AK model considers human capital, like physical capital, as an important factor of production. It assumes that physical capital and human capital can be perfectly substituted in the production process, and $K = H$. So, we can write

the production function as:

$$Y = AK^\alpha H^{1-\alpha}, \quad A: \text{technology factor}, \quad H: \text{human capital}, \quad K: \text{physical capital}$$

Due to the perfect substituted between K and H , we can rewrite the function as:

$$Y = AK$$

This is so called "AK production function". We can use the same way to derive the fundamental accumulation equation

$$\dot{K} = sAK - \delta K$$

divided by K in both side:

$$\frac{\dot{K}}{K} = sA - \delta$$

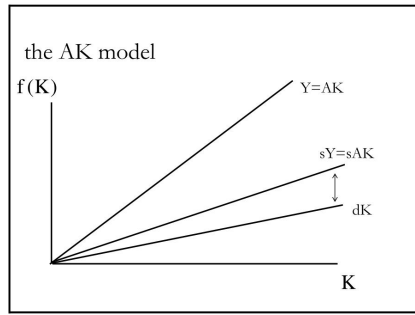


Figure 1: AK Model

In the diagram, dK is δK . We can see that sAK is always greater than δK , so capital will continue to accumulate. As capital accumulates, the next period's capital will continue to increase, so each period's output and capital per worker will continue to grow, indicating that $\dot{K} > 0$. This is how the AK production function can reasonably explain the fact of income per worker growth.